

Clear. The check is complete.

How clear is the choice?

Clear

Close call

No good option

Clear

One option is better on every measure that mattered.

Based on 1 option: the winner leads on every criterion by more than the uncertainty.

How complete is the check?

Incomplete

Mostly

Complete

Complete

The planned review is complete. Its scope and limitations are documented below.

What we need from you

Approve the bounded buffer so the build can start.

Effort: Small

Why: It is the only design where a full buffer tells the sender to retry instead of losing events.

What was checked

Bounded buffer	Checked
Spill to disk	Checked
Priority shedding	Checked

1 option needs attention; the working summary on the next page lists it.

Summary

Option A — bounded buffer with early rejection — is the only shape whose failure mode reaches the producer.

Three approaches to shedding load on the ingest queue were drafted independently and judged against one brief. Option A — a bounded buffer that rejects early — is the only shape whose failure mode reaches the producer, and it is recommended. The ring's size is left open and should be measured rather than guessed.

Why this rating

Option	Meets the goal	Leads on every criterion	Lead beyond uncertainty	Level	Answered by
1. Bounded buffer that rejects early	Yes	Yes	Yes	Clear	Closeout

Needs attention

	Status	Option	Where
1	Selected	Bounded buffer that rejects early	internal/ingest/admit.go:42

Examined and sound

Finding	What was sound
2	Spill overflow to disk
3	Shed by producer-declared priority

Scope and limits

- No load test was run; the 200ms target is unverified against any of the three shapes.
- The ring size is left open and should be derived from a measured drain rate.

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Options

1. Bounded buffer that rejects early

Needs action **Selected** **A**

Impact The only shape whose failure mode reaches the producer: a full ring returns 429 and the producer keeps the event.

Action Adopt. The producer already retries transport errors, so the added cost is a status-code branch.

Where internal/ingest/admit.go:42

2. Spill overflow to disk

Sound

Rejected

B

Impact Moves the bound to the disk rather than removing it, and exhaustion there is silent and unrecoverable without an operator.

Action Revisit only for a future producer that cannot retry, never as the default.

Where docs/ingest-design.md:142

3. Shed by producer-declared priority

Sound

Rejected

C

Impact Correctness would depend on a classification the receiver cannot verify, turning a load problem into silent data loss.

Action Do not adopt.

Where internal/ingest/classify.go:31

Assessment

Options: how the ingest queue should shed load

Three approaches were drafted independently and judged against the same brief: keep p99 admission latency under 200ms when the upstream producer bursts to ten times its steady rate, without dropping an event the producer believes it delivered.

The options as drafted

Option

A — Bounded buffer, reject early

Shape A fixed in-memory ring in front of the writer. When the ring is full the admission handler returns 429 with a retry hint, so the producer keeps the event and retries it.

What it costs The producer must implement retry with backoff. Nothing else changes, and the failure mode is one the producer already handles for network errors.

B — Spill to disk

Shape Overflow is written to a local segment file and drained when the writer catches up. Admission never fails while disk remains.

What it costs A second durability story: segments must be fsynced, replayed on restart, and garbage collected. Disk-full becomes a new failure mode with no back-pressure signal.

C — Shed by priority

Shape Events carry a class; under pressure the low class is dropped and the high class is admitted.

What it costs Requires the producer to classify every event correctly, and the classification becomes a contract the receiver cannot verify. Silent loss is invisible to the producer.

Why A

Option A is the only one whose failure mode the producer can observe. B moves the problem rather than solving it — the queue still has a bound, it is just the disk, and reaching it is now silent and unrecoverable. C makes correctness depend on a classification the receiving side cannot check, which turns a load problem into a data-integrity problem.

A's cost is real: the producer needs retry with backoff. That work is bounded, it is the producer's own code, and it makes back-pressure explicit rather than hiding it behind a buffer that will eventually fill anyway.

What this does not settle

- The ring's size is left open. It should be derived from the measured drain rate rather than guessed.

- If a future producer cannot retry, B becomes worth revisiting for that path alone — not for the default.

Adjudication

Judge

Selected

Judging the drafted options · recorded 14 July 2026, 09:19 UTC

Winner A — bounded buffer with early rejection

Rationale

A is the only option whose failure mode reaches the producer. B replaces a bounded queue with a larger bounded queue and adds a silent, unrecoverable failure at the new bound; C makes correctness depend on a classification the receiver cannot verify. A's cost — retry with backoff at the producer — is bounded, local, and already implemented for network errors.

Claim	Statement
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judge-1	Option B does not remove the bound, it moves it to the disk, where exhaustion produces no back-pressure signal and cannot be recovered without operator action. docs/ingest-design.md:142
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judge-2	Option C's priority classification is set by the producer and cannot be validated by the consumer, so a misclassification presents as silent data loss rather than as an error. internal/ingest/classify.go:31
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judge-3	The producer already implements retry with exponential backoff for transport errors, so option A's added cost is a status-code branch rather than new machinery. grep#7
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Evidence appendix

3 records · all cited · 2 read, 1 grep

2 Read records · all structured

Handle	Location	Cited by
read#1	docs/ingest-design.md:140-144 140 ## Durability 141 142 Segments are fsynced on rotation only; a crash between rotations loses the tail of the open segment.	report, judge-1
read#2	internal/ingest/classify.go:31 31 // Class is set by the producer and is not validated here.	report, judge-2

1 Grep record · all cited by judge-3 · all structured

Handle	Location	Matches
grep#7	internal/ producer retry.go:58: backoff := b.Next()	9 matches in 2 files

About this document

Produced by Painted Wolf Code, paintedwolf.ai
Workflow options 1.2.0
Run b71e4a02-5c88-4d13-96af-2e0f7d6b4c95
Project acme/app
Head 9f8e7d6c5b4a39281706f5e4d3c2b1a098765432
Started 14 July 2026, 08:51 UTC
Completed 14 July 2026, 09:22 UTC
Model claude-opus-5 via anthropic
Workers 4 legs: option-drafter ×3, judge ×1
Evidence 3 records observed

This document is a projection of the run's durable records — its closeout, review verdicts, scan findings, evidence ledger, and visuals — rendered deterministically: the same records produce byte-identical output. The records, not the document, are the record of the outcome.